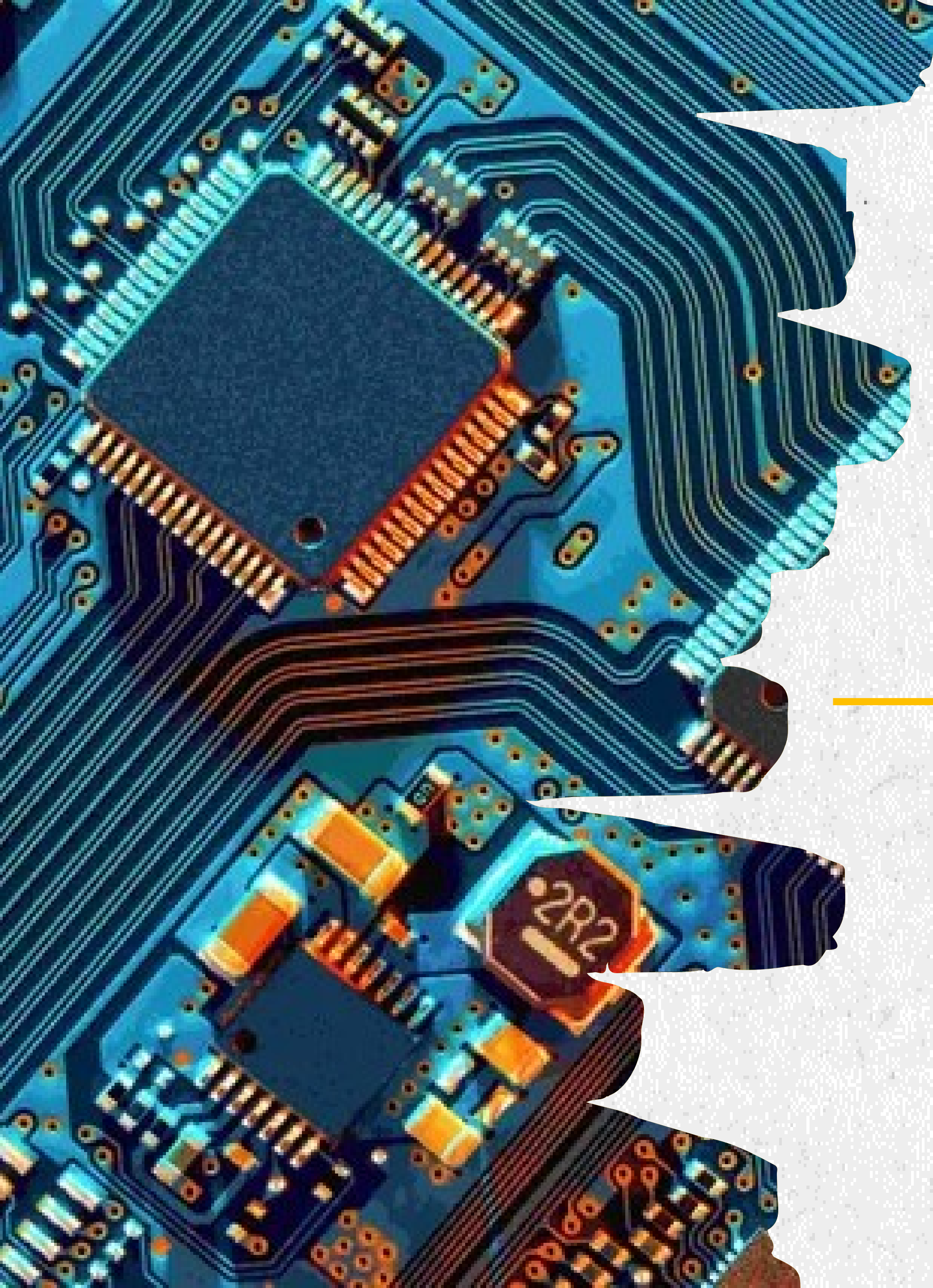
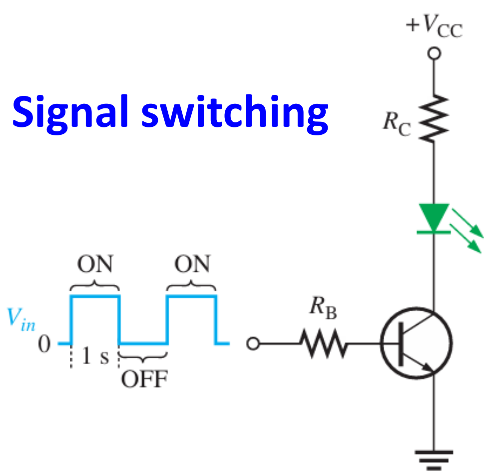


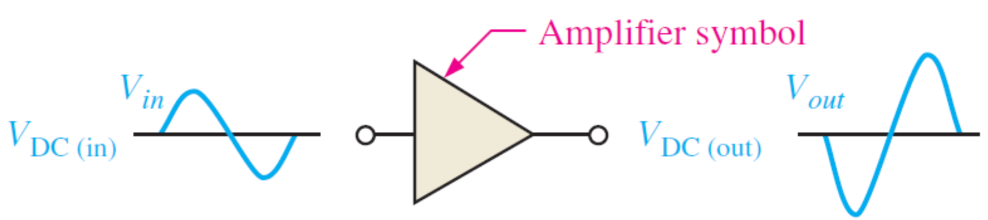
CHAPTER 7
BJT AMPLIFIERS

BJT AMPLIFIERS

- As reviewed in the previous Chapter, BJTs can be applied for two basic applications:
  1. **Signal switching 訊號開關** – to use a small signal to control BJTs as a switch: on (saturation)/off (cutoff).
  2. **Signal amplification 訊號放大** – to *amplify a small signal into a magnified signal* at the output port. This will be discussed more details in this Chapter.

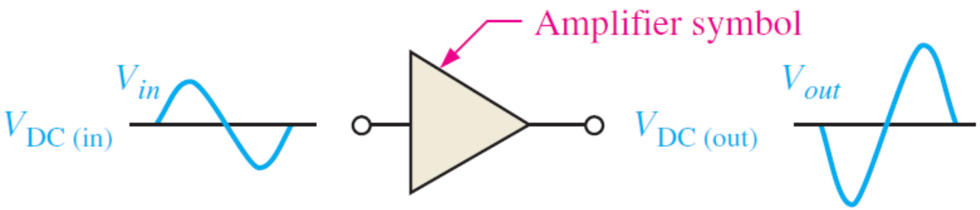
Signal switching





Signal amplification

VOLTAGE-DIVIDER BIAS

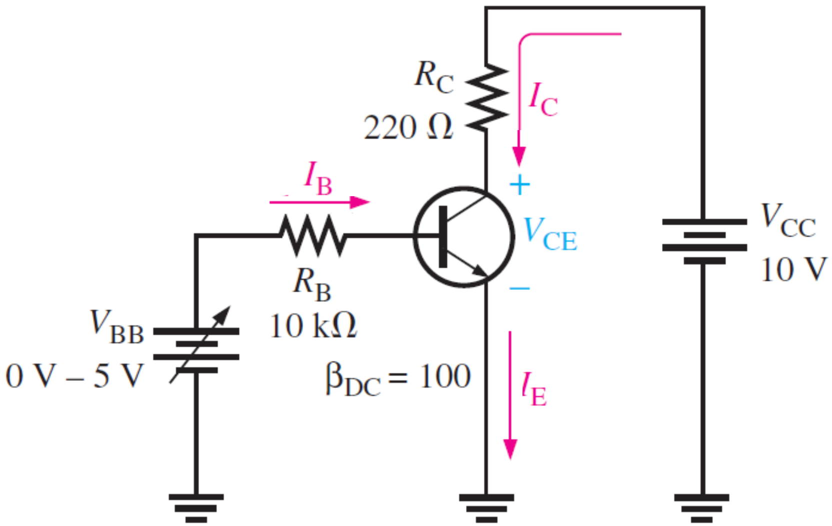


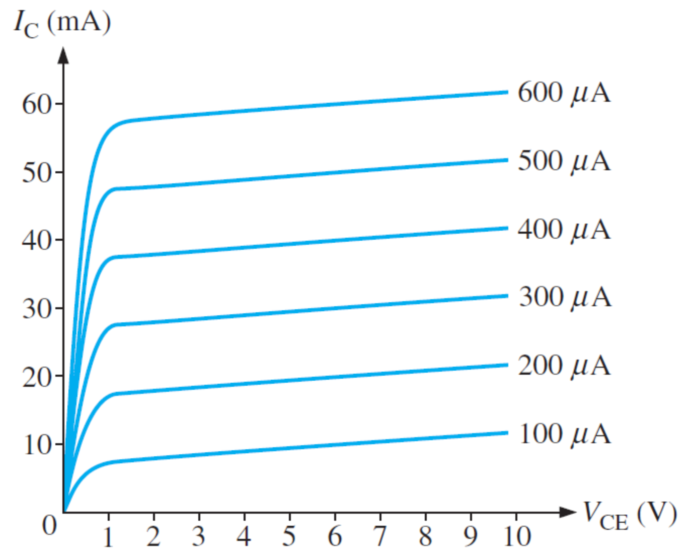
Signal amplification

- BJTs work very well with the *positive input signal* (forward biased), while the *negative signal is **blocked** due to the **reverse bias***.
- To amplify the signal by BJTs,
  1. Add an DC offset to the AC input signal, such that signal is all positive!
  2. Amplify the (DC+AC) signal
  3. Filter out the DC signal, remaining the amplified AC signal

➢ ***Voltage-divider* (電壓分配) *bias*** – how to add up and analyze the DC offset.

DC LOAD LINE





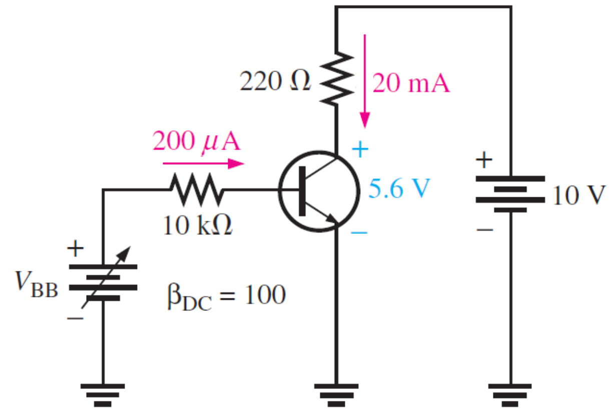
- Recall the basic BJT circuit, let us fix V_{CC} and vary V_{BB}.

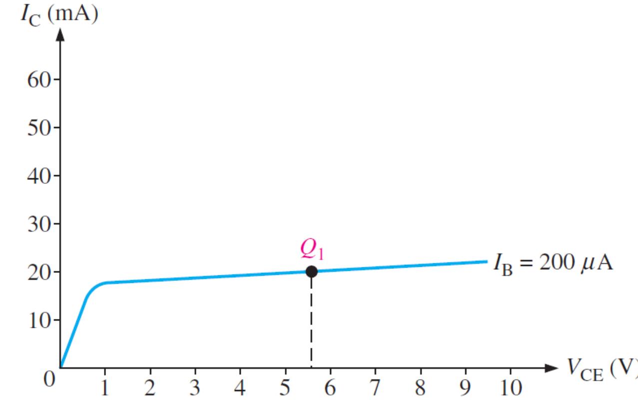
Set V_{BB} such that $I_B = 200\ \mu A$

$$I_C = \beta_{DC} I_B = 20 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C(220)$$

$$V_{CE} = 5.6 \text{ V}$$



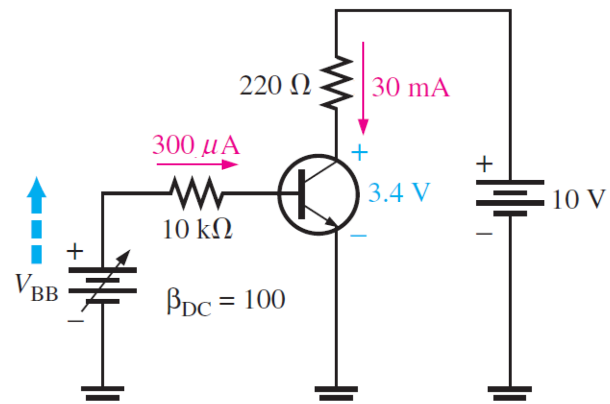


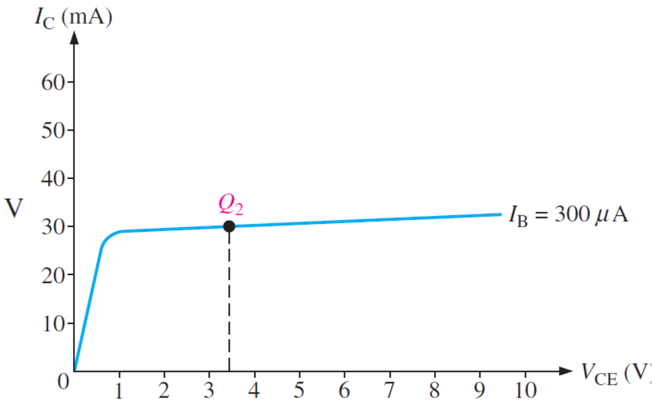
Set V_{BB} such that $I_B = 300\ \mu A$

$$I_C = \beta_{DC} I_B = 30 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C(220)$$

$$V_{CE} = 3.4 \text{ V}$$



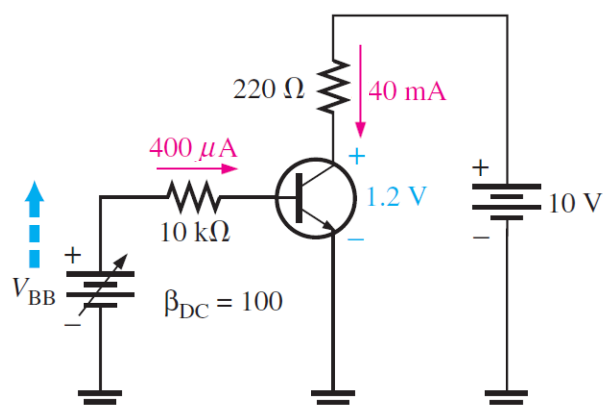


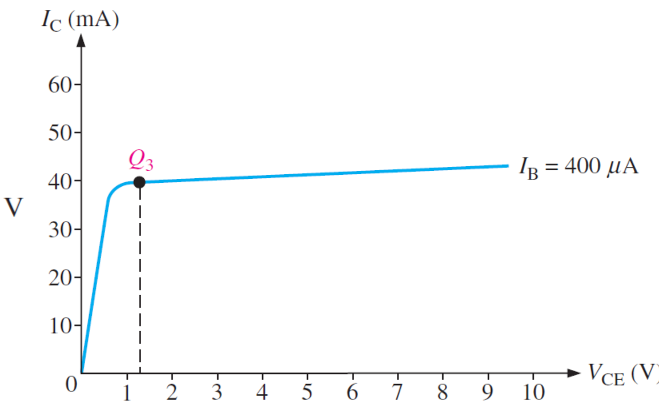
Set V_{BB} such that $I_B = 400\ \mu A$

$$I_C = \beta_{DC} I_B = 20 \text{ mA}$$

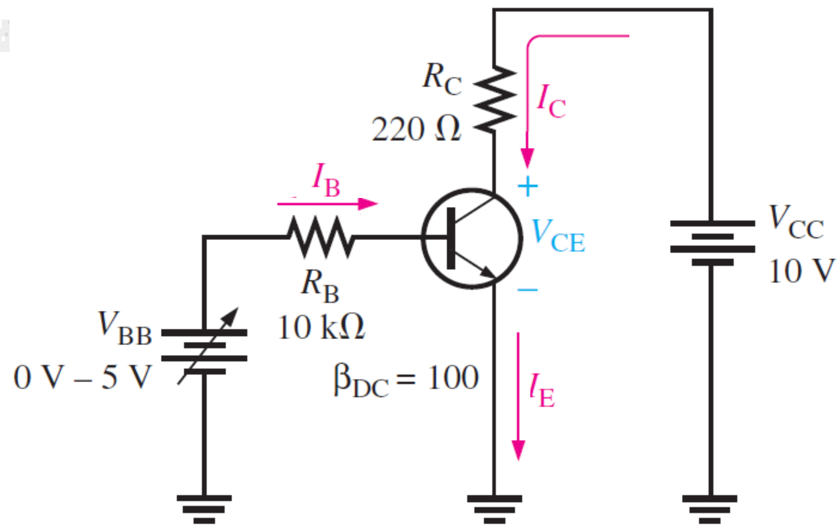
$$V_{CE} = V_{CC} - I_C(220)$$

$$V_{CE} = 1.2 \text{ V}$$

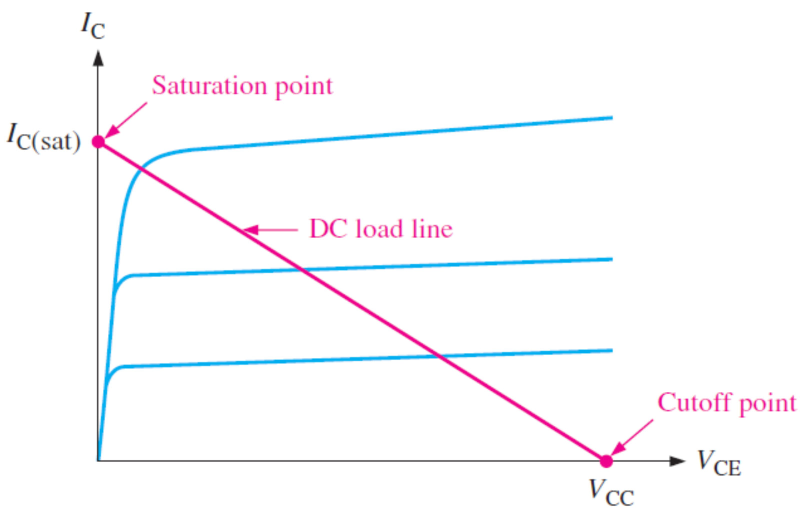




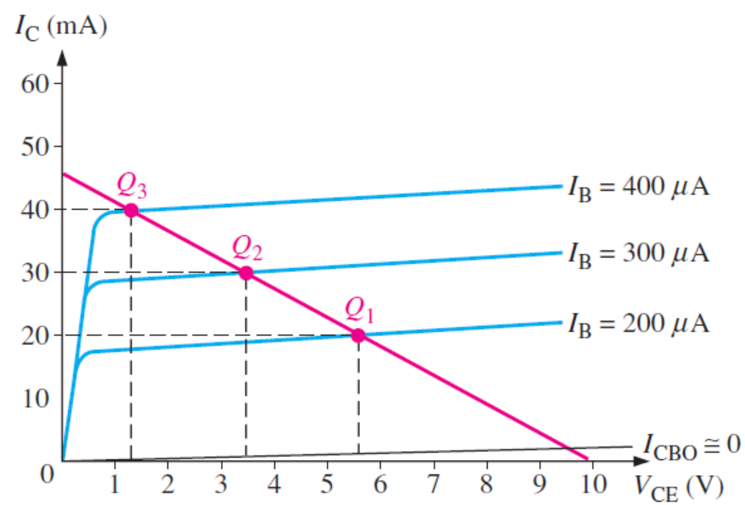
DC LOAD LINE



R_C
220 Ω
I_C
I_B
+
V_{CE}
−
V_{CC}
10 V
R_B
10 kΩ
V_{BB}
0 V – 5 V
$\beta_{DC} = 100$
I_E



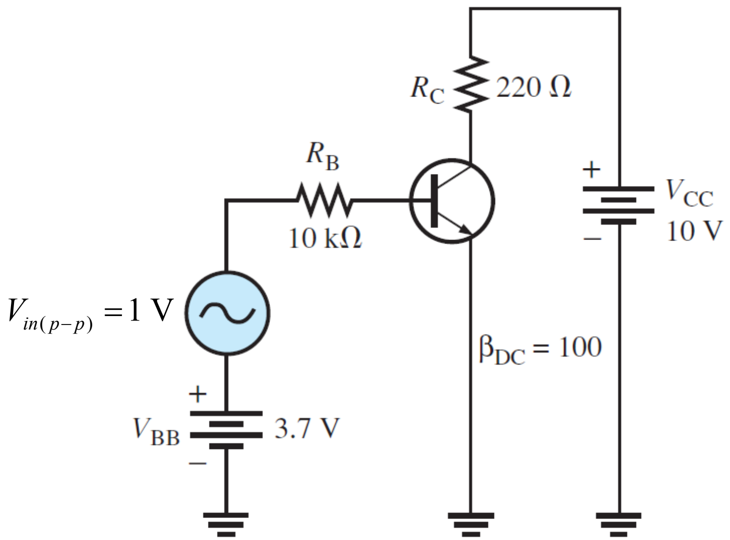
I_C
Saturation point
$I_{C(sat)}$
DC load line
Cutoff point
V_{CE}
V_{CC}

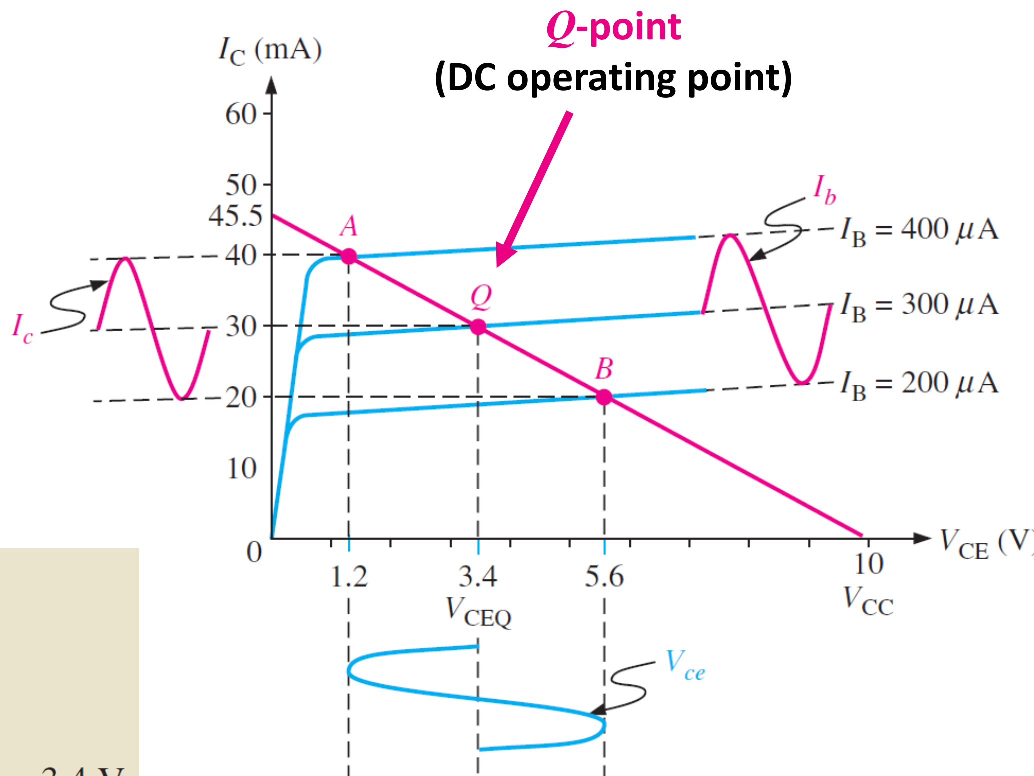


I_C (mA)
60
50
40
30
20
10
0
Q_3
Q_2
Q_1
$I_B = 400\ \mu A$
$I_B = 300\ \mu A$
$I_B = 200\ \mu A$
$I_{CBO} \cong 0$
1 2 3 4 5 6 7 8 9 10
V_{CE} (V)

BJT WITH AC SIGNAL

- Let us add an AC signal on the DC signal V_{BB}.



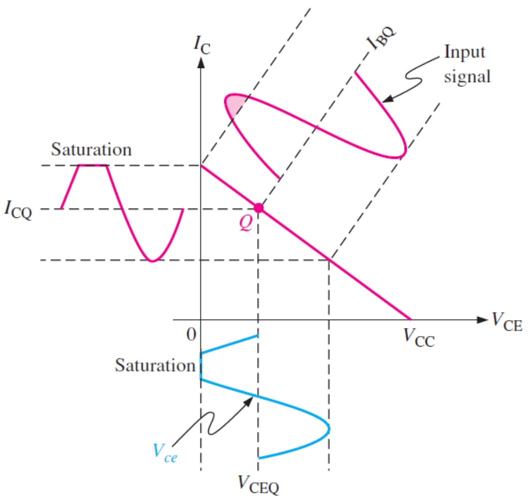


$$I_{BQ} = \frac{V_{BB} - 0.7\text{ V}}{R_B} = \frac{3.7\text{ V} - 0.7\text{ V}}{10\text{ k}\Omega} = 300\ \mu\text{A}$$

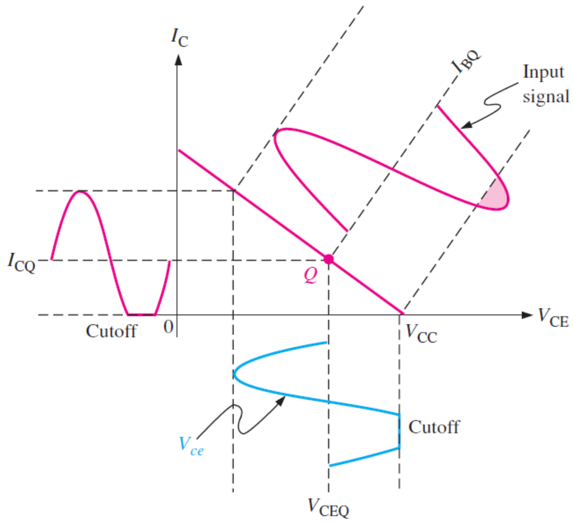
$$I_{CQ} = \beta_{DC} I_{BQ} = (100)(300\ \mu\text{A}) = 30\text{ mA}$$

$$V_{CEQ} = V_{CC} - I_{CQ} R_C = 10\text{ V} - (30\text{ mA})(220\ \Omega) = 3.4\text{ V}$$

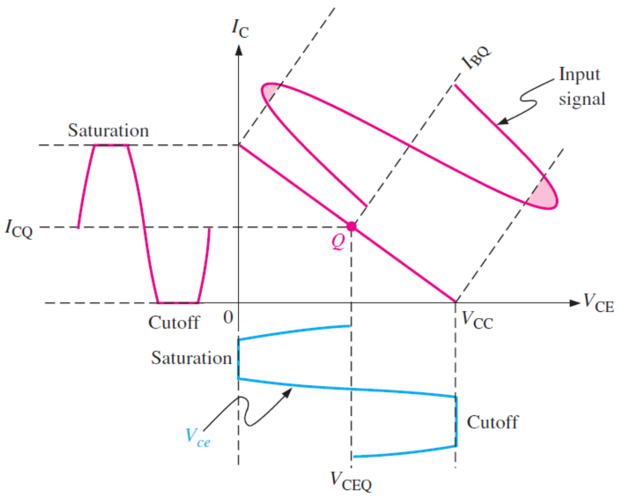
WAVEFORM DISTORTION



Q-point is too close to saturation.



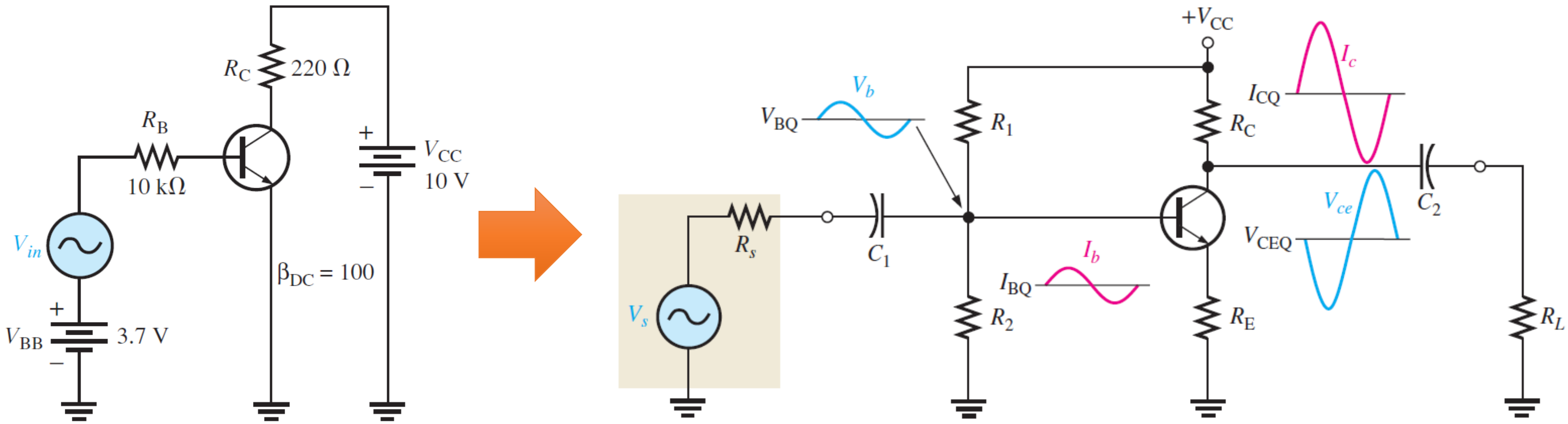
Q-point is too close to cutoff.



Input signal is too large.

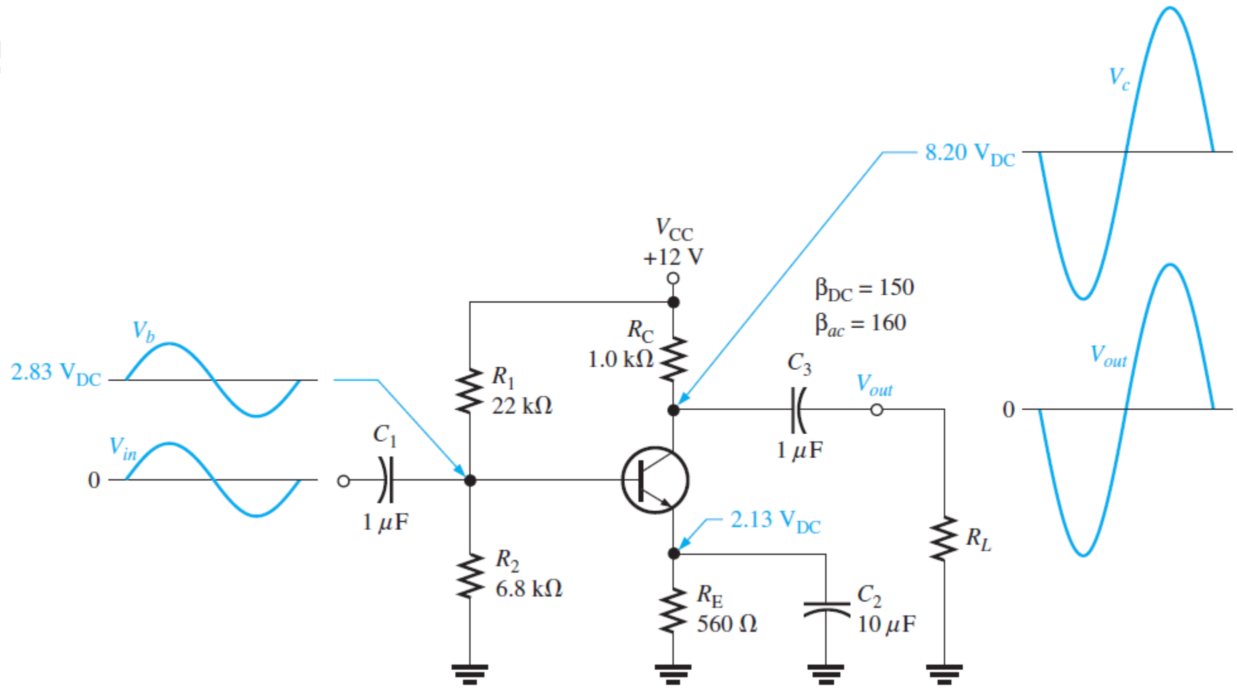
"These three reasons make the amplified waveform ***clipped****."*

BJT AMPLIFIERS FOR AC SIGNALS



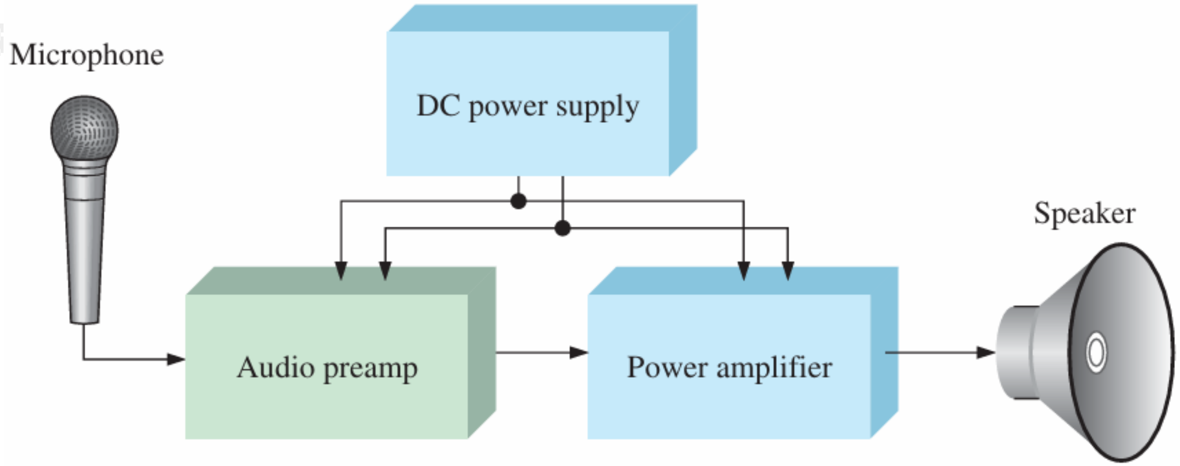
- V_s is the *input AC signal to be amplified*.
- V_{CC} is the one and only DC voltage supply.
- Capacitors C_1 and C_2 are to block DC currents <u>not to flow to V_s and R_L sides</u>, simplifying the circuit analysis.

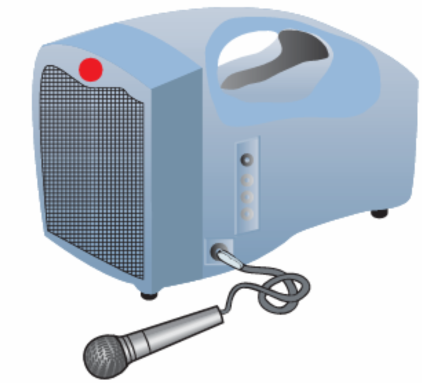
ANALYSIS OF BJT AMPLIFIERS

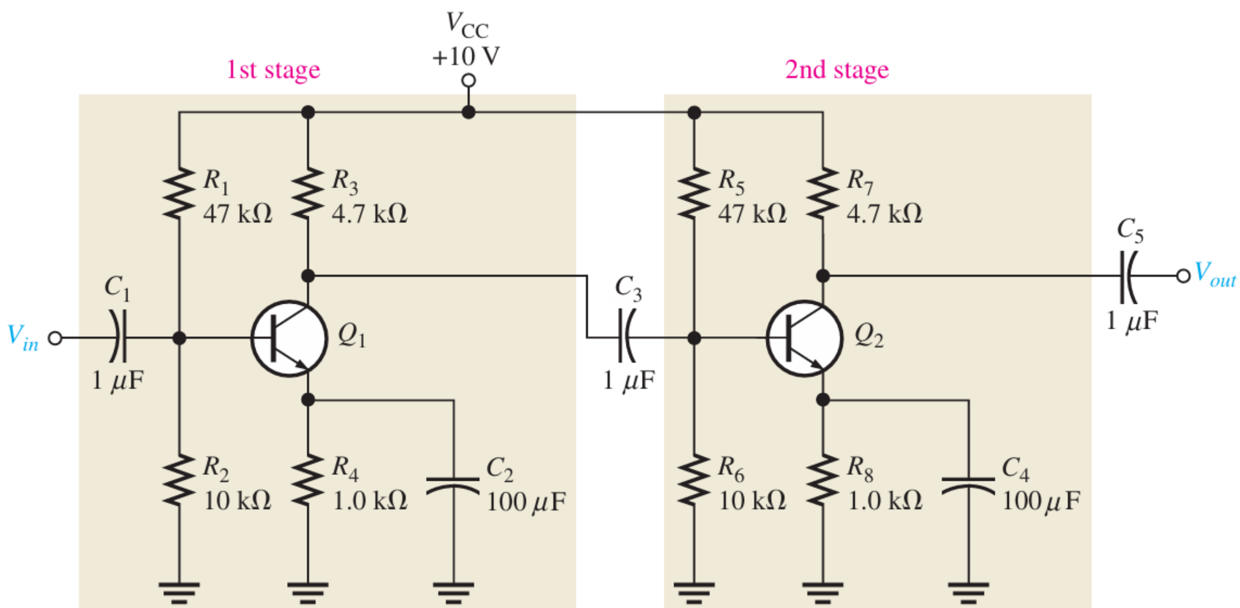


- To analyze this BJT amplifier, we split into 2 analyses:
  1. DC analysis
  2. AC analysis

APPLICATIONS







$\beta_{DC} = \beta_{ac} = 150$ for Q_1 and Q_2